A CLOSED SUBSPACE OF A W-LOCAL SPECTRAL SPACE IS W-LOCAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the identification of the closed point set of the closed subspace with the trace of the closed point set of the ambient space, and the verification that the unique specialization of a point of the subspace remains inside it.

Editorial note. The label lemma-closed-subspace-w-local names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-closed-subspace-w-local –start 419 –end 431 –local-id 28011 returned RESULT=OK (L28011 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfB.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Definition of a w-local spectral space. Same chapter, the definition of a w-local spectral space through its set of closed points and the specialization behaviour. Supplies the two properties to be verified for the closed subspace: the closed points form a closed set and every point specializes to a unique closed point.

Lemma 0.1. *Let X be a w-local spectral space. If $Y \subset X$ is closed, then Y is w-local.*

Proof. The subset $Y_0 \subset Y$ of closed points is closed because $Y_0 = X_0 \cap Y$. Since X is w-local, every $y \in Y$ specializes to a unique point of X_0. This specialization is in Y, and hence also in Y_0, because $\overline{\{y\}} \subset Y$. In conclusion, Y is w-local. □

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